As part of snapshotting, all indexes, all data streams, and the entire cluster state are backed up. Note that after the first snapshot is performed, subsequent backups will be incremental updates, not full copies. There are two ways we can work with snapshots:

- Snapshot and restore RESTful APIs
- The Kibana snapshot and restore feature

The first step is to choose a repository type and register it. Let's use both approaches to register a snapshot repository, as discussed in the next section.

14.4.2 Registering a snapshot repository

To keep things simple, let's pick the filesystem as our repository type: we want to store our snapshots on a disk mounted on a shared filesystem. We begin by mounting the filesystem with available memory on all the master and data nodes in the cluster. Once the server has this filesystem mounted, we need to let Elasticsearch know its location by specifying that in the configuration file.

Edit the elasticsearch.yml configuration file to amend the `path.repo` property, pointing it to the location of the mount. For example, if the mount path is /volumes/es_snapshots, `path.repo` looks like this: `path.repo: /volumes/es_snapshots`. After adding the mount path, we need to restart the respective nodes for this mount to be available for the nodes.

Registering the repository using snapshot APIs

When the nodes are back online after the restart, the final step is to invoke the snapshot repository API. The following listing shows the code.

Listing 14.4 Registering a filesystem-based snapshot repository

```
PUT _snapshot/es_cluster_snapshot_repository
{
  "type": "fs",
  "settings": {
    "location": "/volumes/es_snapshots"
  }
}
```

Names the repository provided to the _snapshot API endpoint

Sets the type of repository to filesystem ("fs")

Defines the location of the repository as a mounted filesystem

Elasticsearch provides the `_snapshot` API for carrying out actions related to snapshots and restores. In listing 14.4, we create a snapshot repository called `es_cluster_snapshot_repository`. The body of the request expects the type of repository we are creating and the properties required to set the repository type. In our example, we set `"fs"` (for filesystem) as our repository type and provide the filesystem path as the `"location"` in the `settings` object.

Because we've already added the mount point in the configuration file and, of course, restarted the node, the code in listing 14.4 should execute successfully to register our first repository. Issuing a `GET _snapshot` command returns the registered snapshot: